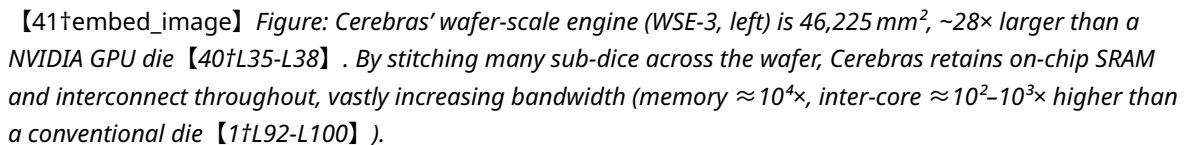


Executive Summary

AI workloads demand ever-growing compute and data-movement capacity. Larger chip dies – or multi-die assemblies – offer a path to pack more cores and on-chip memory tightly, dramatically boosting bandwidth and reducing off-chip traffic [1†L71-L80] [6†L51-L55]. State-of-the-art examples include wafer-scale engines (e.g. Cerebras WSE-3, 46,225 mm² [40†L35-L38]) and chiplet modules (AMD MI300, Graphcore Bow). However, pushing size also stresses lithography limits (reticle ~26×33 mm), yield, power delivery, cooling, and cost. We review manufacturing approaches (stitching, multi-die masks, wafer/panel-scale processing), advanced packaging (EMIB, silicon interposers, CXL, in-package optics), and system challenges (thermal, power, yield, reliability). Key findings: larger integration can yield 100–1000× more inter-core bandwidth than monolithic dies [1†L92-L100], but requires new technologies. Reticle stitching and chiplet tiling let designers “cheat” the size limit [6†L51-L55]

[13†L250-L258]. Panel-level substrates promise even larger “prints” for future AI accelerators [13†L271-L274], though mechanical stresses pose major challenges [30†L168-L177] [30†L190-L198]. Packaging interconnects (EMIB, CoWoS interposers, UCIe optical links) are vital to link multiple tiles with high bandwidth [17†L123-L129] [24†L99-L104]. Thermal/power management (liquid cooling, advanced PDNs) becomes critical beyond a few hundred watts [6†L38-L43] [36†L219-L227]. Yield losses from large die can be mitigated by redundancy (wafer-scale meshes) or chiplet granularity [6†L62-L68] [8†L41-L44]. Cost and supply-chain effects are complex: larger formats amortize fixed costs over fewer units, but panel or wafer-scale methods increase substrate and process costs [30†L125-L133] [8†L41-L44]. We conclude with comparative trade-off tables and recommendations: chiplets and 2.5D packaging are practical today, while wafer- and panel-scale integration are emerging paths (see Table below).

 *Figure: Cerebras' wafer-scale engine (WSE-3, left) is 46,225 mm², ~28× larger than a NVIDIA GPU die [40†L35-L38]. By stitching many sub-dice across the wafer, Cerebras retains on-chip SRAM and interconnect throughout, vastly increasing bandwidth (memory ≈ 10⁴×, inter-core ≈ 10²–10³× higher than a conventional die [1†L92-L100]).*

Background & Motivation

AI scaling laws reward ever-larger models and compute: each generation of LLMs grows in size to improve accuracy [1†L41-L50]. Larger chips can host more AI cores and SRAM, enabling more data to stay on-chip and reducing costly off-chip communication [1†L47-L54]. For inference, for example, system-on-wafer architectures can offer 100–1000× the inter-chip communication and memory bandwidth of monolithic dies [1†L92-L100]. TSMC and IEEE explicitly anticipate a wave of wafer-scale computing for AI [1†L79-L84]. In short, **more silicon area ⇒ more cores + memory + local bandwidth**, which for memory-bound AI is a direct efficiency and performance win [1†L71-L80].

However, semiconductor manufacturing imposes hard limits on die size. The maximum reticle field on today's lithography steppers is ~26 mm×33 mm (~858 mm²) [6†L45-L53], so a conventional single-die chip is limited to ≲900 mm² (Cerebras notes NVIDIA's largest dies are ~800–850 mm² [6†L45-L53]). Thus “bigger” generally means going **beyond a single reticle field** by stitching or tiling chips. This is why major AI processors now use 2–4 large tiles (e.g. NVIDIA's Blackwell GPUs are two-reticle packs [13†L256-L264]) or wafer-scale assemblies (Cerebras, Graphcore). The payoff is enormous: for example, Cerebras WSE-3's 46,225 mm² wafer core hosts ~4 trillion transistors and 900,000 AI cores

【40†L35-L38】 , enabling on-chip memory of tens of GB and inferred performance of 125 PFLOPS. In comparison, a state-of-the-art monolithic GPU or APU holds <100 GB of HBM with <10s TB/s bandwidth.

Current State-of-the-Art

- **Largest dies/chips:** Cerebras WSE-3 (2023) is a full 300 mm wafer reticle-stitching assembly of ~46,225 mm² 【40†L35-L38】 . Graphcore's Bow IPU (2022) uses *wafer-on-wafer* stacking (two 7nm wafers bonded) to improve power delivery 【15†L224-L230】 . AMD's MI300A (2023) is a 4-chiplet APU with 5.3 TB/s HBM and a chiplet mesh fabric. NVIDIA's Blackwell GPUs (~2024–25) use 800 mm² dies in 2-or-4-tile packages. These examples illustrate chip-scale vs package-scale scaling strategies: Cerebras pushes a single package across a wafer, Graphcore stacks wafers, AMD/NVIDIA tile multiple silicon pieces in a package.
- **Wafer sizes and reticle limits:** Most fabs use 300 mm wafers; 450 mm efforts stalled. Reticle fields at 300 mm are ~26×33 mm 【8†L96-L104】 . Standard photomasks are ~104×132 mm, projecting a 26×33 mm field onto wafer 【8†L96-L104】 . Full use of the reticle (26×33) means 12×10 = 120 exposures per wafer 【8†L118-L124】 . Designs near this size maximize throughput. Exceeding reticle size requires stitching (see below). Note that waste ("scribe" edges) on round wafers grows as chips become large and rectangular 【13†L252-L261】 ; e.g. ~800 mm² square chips leave ~20–30% of a 300 mm wafer unused 【13†L256-L264】 .
- **Stitching and multi-die reticles:** To surpass the lithography field, some projects use *stitching*: exposing multiple adjacent fields and aligning them with nanometer precision. Cerebras' WSE adds an extra metal layer for "cross-die wires" to connect adjacent reticle tiles 【6†L51-L55】 . In essence, WSE is composed of many ~860 mm² tiles that are *almost* separate chips but joined on-wafer by stitching 【6†L51-L55】 【2†L29-L33】 . This yields a contiguous compute fabric over the wafer (reducing inter-chip latency to on-chip levels) while still honoring reticle constraints. Another approach is multi-die masks: placing multiple distinct die patterns per reticle. Intel's Foveros and Nvidia's NVLink-SLI use separate die but standard packaging (no wafer stitching).
- **Chiplet and interposer (2.5D):** Major products now use chiplets on silicon interposers or in packages. For example, AMD's EPYC and Instinct series tile CPU/GPU chiplets on a large silicon interposer (TSMC CoWoS) with HBM on the same interposer. Intel's EMIB (Embedded Multi-die Interconnect Bridge) embeds small silicon "bridges" between large dies in a package 【17†L123-L129】 . EMIB uses tiny interconnect bridges only where needed (saving silicon area) 【17†L123-L129】 . 2.5D interposers provide ultra-high I/O (TSV-enabled) but are expensive; EMIB and similar *silicon bridge* approaches trade some density for lower cost 【17†L123-L129】 【17†L99-L107】 .
- **3D stacking:** Face-to-face or face-to-back die stacking (hybrid bonding) is emerging for heterogeneous integration. For example, 3D-ICs stack logic atop RAM (like HBM-stacked DRAM or cache-on-logic (Milan-X)). Graphcore's Bow was a wafer-on-wafer 3D stack dedicated to power delivery 【15†L224-L230】 . Intel's Foveros (and AMD's 3D V-Cache) stack dies vertically. These raise bandwidth further and shorten certain paths (e.g. logic-to-cache), but thermal and yield become more challenging.

【34†embed_image】 *Figure: Steps to integrate a small silicon bridge (EMIB) in an organic package* 【17†L123-L129】 . EMIB embeds a tiny silicon "interposer" in the substrate, connecting multiple dies via fine-pitch bumps. Compared to a full silicon interposer, EMIB uses silicon only under the bridge area, reducing cost 【17†L123-L129】 .

Manufacturing Larger Dies

- **Reticle stitching:** Advanced lithography scanners can stitch adjacent fields. ASML's high-NA EUV tools still have a 26 mm slit, but wafer-stage motion allows contiguous exposures. Precise alignment enables essentially “drawing” a larger layout by multiple steps [6†L51-L55] [8†L96-L104] . Cerebras proved it: they add *cross-die metal routing* above the standard layers so that circuits on separate reticle fields electrically connect [6†L51-L55] . Technical challenges include alignment tolerances (μm -level over 200+ mm) and overlay. Stitching increases mask and scan overhead (if many reticle fields are stitched, yield and throughput suffer). But it permits “monolithic” die logic beyond 1x field.
- **Multi-die reticles:** Instead of stitching a single design, designers may pack multiple chips on one reticle. For example, AMD's MI300 features several identical 370 mm² logic tiles per wafer. The reticle can contain more than one die image, improving reticle-utilization [8†L98-L107] . However, if dies are large enough to need single placement (as in the example [8†L109-L113]), throughput drops because each exposure prints fewer chips.
- **Wafer-scale integration (WSI):** WSI aims to use the entire wafer as *one big “chip”*. True WSI uses wafer-size masks or stitching. Cerebras' WSE-3 is WSI via stitching: it treats many reticles as sub-chips and has redundant cores/links to bypass defects [6†L64-L68] . Yield is managed by a “fail-in-place” design: spare rows of cores and routers allow the fabric to reconfigure around faults [6†L62-L68] . WSI can realize extremely high compute density (millions of cores, 10s of TB/s on-chip bandwidth [1†L71-L79]), but at cost: manufacturing yield can be as low as a few percent without redundancy [6†L57-L65] , and retesting such a large chip is non-trivial.
- **Panel-scale processing:** Going beyond 300 mm wafers, panel-level packaging (PLP) uses large rectangular glass or silicon panels (e.g. 300×300 mm or up to 600×600 mm) for lithography and assembly [13†L271-L274] . This is driven by AI: round wafers can only fit a few huge rectangular chips (wasting area) [13†L252-L261] . Lam Research predicts panels will take off when reticle area $\approx 4,500\text{--}7,700\text{ mm}^2$ (around 2027–2030) [13†L271-L274] . Panel benefits include many more chips per substrate (9× productivity is cited for a 600 mm panel vs 300 mm wafer [12†L321-L324]) and symmetric thermal distribution. However, panels introduce severe mechanical challenges: glass (typical substrate) is brittle, micro-cracks and warpage are amplified at large area [30†L168-L177] [30†L190-L198] , and process steps (etching, via-fill) are harder to uniformize. Panel processes (lithography, via drilling) must be redesigned for larger fields. Panel-scale fabrication is still in pilot phase (Lam sees low-volume RF chip production already, but AI accelerators are still development [13†L279-L284]). The trade-off is future capacity vs current risk: panels could enable enormous “mega-chips”, but require new supply chains (glass substrates, maskless lithography [12†L321-L324]).
- **Large reticle lithography:** There is R&D on larger steppers with $\sim 600\text{ mm}^2$ fields (e.g. Nikon DSP-100 supports up to 600 mm substrates [12†L307-L314]). These tools could print wide slabs (useful for large interposers or fan-out packaging), but are not yet mainstream for logic. High-NA EUV (launching $\sim 2024\text{--}25$) maintains $\sim 26\text{ mm}$ slit but halves scan length [8†L129-L137] , improving resolution but not reticle size. Ultimately, reticle area will grow only modestly (Lam forecast $\sim 7,700\text{ mm}^2$ by 2030 [13†L271-L274]) before plateau, keeping pressure on multi-die solutions.

Packaging and Interconnect Options

- **2.5D interposers (CoWoS/InFO):** Silicon interposers allow side-by-side die placement with ultra-dense wiring and TSVs. AMD GPUs/CPUs (e.g. MI300) use a large CoWoS interposer connecting multiple compute tiles and HBM stacks. Advantages: very high bandwidth (HBM links, dense mesh), close proximity of chips, proven high-end use. Drawbacks: silicon interposers are expensive ($\geq \$1000$ each for large ones) and limited in size (current $\sim 600 \times 600 \text{ mm}^2$ projects). Interposers also carry thermal issues as heat must pass through silicon.
- **EMIB and silicon bridges:** Intel's EMIB embeds small silicon "bridges" (few mm^2) in the substrate between chips [17†L123-L129]. EMIB provides many TSV-like links (e.g. $55 \mu\text{m}$ bump pitch, hundreds of wirings) [32†L216-L224], comparable to interposers in bandwidth, but cheaper because only tiny bridge pieces are silicon; the rest of the package is organic. SemiEngineering notes EMIB can deliver "fairly similar bandwidth" to interposers [17†L123-L129]. EMIB is flexible (can connect chips of different sizes/nodes) and easier to manufacture (no full-wafer silicon needed), but it places thermal expansion stress on package adhesives and requires very planar assembly.
- **CXL and PCIe interconnect:** Compute Express Link (CXL) is an emerging CPU-centric interconnect for coherent memory and I/O pooling. While not a chip-package technology per se, CXL (over PCIe 5/6/7) enables attaching disaggregated memory or accelerators over the PCIe bus with coherency. CXL 3.0 (2023) and 4.0 (2025) double bandwidth: CXL 4.0 (PCIe 7.0 base) achieves 128 GT/s ($\sim 1.5 \text{ TB/s}$ with bundled ports [43†L90-L98] [43†L132-L139]). This is an order-of-magnitude higher throughput than PCIe 5 or UCIe copper links. CXL is important for data-centre architecture: it allows racks of GPUs/CPUs to share memory coherently, alleviating off-chip memory bottlenecks. It complements on-package solutions by extending the memory domain beyond one node (with $\sim 200\text{--}500 \text{ ns}$ latency [43†L78-L80]). As of 2025, CXL switches and memory devices exist, and CXL is expected in GPUs/CPUs in late-2025 to 2026.
- **Optical interconnects:** Photonics promise low-power, high-bandwidth links over longer distances (mm to meters). Current data centers use optical cables for rack-to-rack (e.g. $100\text{--}800 \text{ Gbps}$ transceivers). On-chip/near-chip optics (co-packaged optics, CPO) are emerging: e.g. AMD's recent CDNA GPUs plan on-package optical for NVLink. The new UCIe chiplet standard explicitly includes an optical interface [24†L47-L54] [24†L99-L104]. Ayar Labs' TeraPHY chiplet (UCIe/AIB compatible) delivers up to 2 Tbps optical links per package [24†L119-L128] ($\approx 64 \times$ PCIe 5 lanes) between packages. Optical links have $< 0.1 \text{ pJ/bit}$ advantage per mm over copper and can span meters [24†L119-L128], enabling multi-shelf composability. However, co-packaged optics add complexity (laser sources, thermo-optic design) and are just reaching early production. For near-term AI chips, optics mainly appear as board-level fabrics (e.g. NVIDIA NVSwitch with optical cables) or planned in-package demos (Intel's Sapphire Rapids had a CPO test). In summary, **UCIe + optics** is a promising path to glue chiplets across a package or board at AI-relevant bandwidth (multi-Tbps) and reach, but is still nascent.

Thermal and Power Delivery

As die or package power grows to hundreds of watts (and beyond 1 kW for multi-chip racks), cooling and PDN design become critical. Large dies have high *power density*; for example, Cerebras' 20 kW WSE-2 required a custom water-cooled chassis [6†L38-L43]. Intel notes today's workloads push "kilowatt-level" power, up from watts decades ago [36†L142-L150]. Key challenges: removing heat

from a large area, managing thermal gradients (differential expansion), and delivering low-impedance power to all cores.

- **Cooling:** Traditional air-cooling hits limits above ~300–400 W per chip. Liquid cooling (cold plates, direct-to-chip water loops) is widely used in AI servers (e.g. NVIDIA H100 SXM, AMD MI300). Even then, tanks of water or glycol are needed for >1 kW nodes. Immersion cooling (chip in dielectric liquid) is another trend. As chips enlarge, uniform cooling is harder; the edges may run hotter than center. Planning heat exchangers and flow is a custom task. Differential CTE (thermal expansion) across a 200+ mm silicon piece can stress packages and require careful mechanical design [6†L38-L43] .
- **Power delivery:** A large GPU or multi-die module may need thousands of amps at ~1 V. This means thick copper planes and many voltage regulators on-board. 3D stacking solutions (like Graphcore's two-wafer Bow) partly address IR drop: Bow stacked a power wafer with deep-trench capacitors under the logic, improving local decoupling [15†L224-L230] . In general, larger chips require *distributed PDNs*: many on-chip regulators, multi-phase VRMs close to each chip, and possibly multi-level supply (48 V drops to ~12 V at board, then to die). Intel highlights that power delivery grows as challenging as data movement [36†L219-L227] . Package designers use techniques like power-grid simulations, epoxy underfill for heat spreading, and copper pillars with filled vias to strengthen PDN.
- **Thermal solutions:** Beyond cooling hardware, advanced materials (slugs, diamond heatsinks) and substrate innovations (glass carriers in panel packaging [30†L168-L177]) help. System-level, racks must manage big PSUs (10–20 kW per node), phase-change cooling, and ensure redundancy (e.g. dual PDN rails, backup chillers).

Yield and Defect Mitigation

Die yield typically follows an exponential drop with area. If defect density is D [defects/cm²], then yield $\approx \exp(-A \cdot D)$ (or more complex models [8†L41-L44]). A ~25×25 mm die might see ~50% yield, whereas a 200×200 mm wafer-scale die would be effectively zero without mitigation. Strategies:

- **Chiplets for yield:** By dividing a large design into smaller chiplets, each die's yield is higher and failures only scrap one tile instead of the whole. Semianalysis examples show 800 mm² yields ~30 chips/wafer vs 4× 200 mm² yields ~79 usable chips [8†L59-L68] . Chiplet modules can bin or disable faulty tiles. The overall product yield improves and wafer productivity increases [8†L59-L68] .
- **Redundancy (WSI):** Cerebras and others build redundancy *on the wafer*. Cerebras adds extra rows/cores and routing links so that if one core/link is defective, it can route around it [6†L62-L68] . Their software remaps around faults so the system “sees” a perfect 2D mesh [6†L62-L68] . This *fail-in-place* approach tolerates many defects (Cerebras claimed production wafers with many nonfunctional cores still work). The trade-off is area overhead (spares) and design complexity. However, it avoids sacrificing an entire large chip to a single defect.
- **Partitioned architectures:** Many large AI packages have redundancy at a higher level. For instance, NVIDIA's multi-GPU stacks (NVLink fabrics) can fall back if one GPU is bad, or data-center clusters deploy many accelerators so that losing one is non-catastrophic. Similarly, AMD chiplet designs often disable a failing tile and sell a lower-tier part.

- **Panel/large-substrate yield:** Glass panel yield is currently low for logic processing. The panel-engineering literature warns of micro-cracks and warpage causing yield loss [30†L168-L177] [30†L190-L198] . Early panel fabs will likely see more than wafer-level scrap rates until processes mature. Hence, WSI and panels rely on different economics (high fixed costs but amortized over few devices) and have yield models that assume redundancy or rework (e.g. testing panels and marking bad regions).
- **Testing:** Testing a huge die or panel is challenging. Burn-in of a wafer-scale chip requires custom test fixtures (the chip itself is the wafer!). Chiplets allow testing each die before assembly (“known good die”). Panel substrates require new wafer-prober tech with larger field of view.

Cost and Supply-Chain Implications

- **Manufacturing cost:** Larger dies mean fewer units per wafer and higher cost per good chip [30†L125-L133] . Even before yield, lithography overhead rises: if a 25×32 mm die uses only 95% of reticle, a larger 13.5×32 mm die (two per reticle) forces more exposures [8†L118-L124] . Packaging costs soar: silicon interposers and silicon bridges involve expensive materials and steps [17†L123-L129] . Panel substrates and related tooling (new scanners, laser debonders) are capital-intensive. However, some savings accrue: chiplets reuse the same masks and can leverage smaller fab runs with better yields. Semianalysis notes chiplets could cut logic-silicon cost ~30% even including additional packaging [8†L59-L68] .
- **System cost trade-offs:** A wafer-scale module like Cerebras’s is few per customer (ultrahigh compute density), so amortized cost per FLOP can be competitive. But it requires custom systems (liquid-cooled racks, special boards), so TCO is different. Chiplet-based GPUs/accelerators can piggyback existing infrastructure (PCIe slots, NVLink cables) which reduces incremental cost. Panel-scale manufacturing, once mature, could reduce cost per chip (many rectangles per panel) and per connectivity (larger interposers on panel rather than one wafer); Yole forecasts PLP revenue growth to \$650M by 2030 driven by AI/HPC [11†L13-L20] .
- **Supply chain:** Wafer- and panel-scale integration increase reliance on specialized suppliers. For example, glass substrate makers (SKC/Absolics) now invest heavily in panel facilities [12†L294-L302] . Foundry demand for large reticle exposure tools will rise (Nikon DSP-100 sales already target 600 mm² substrates [12†L307-L314]). Packaging houses must adapt: CEA Leti, ASE, and others have pilot panel lines [30†L154-L163] . Conversely, if large-die designs fail, thousands of panels/wafer could be wasted—risking large capital. In summary: transitioning to very-large-chip production requires ecosystem alignment (materials, EDA, fab, assembly), and any mismatch can bottleneck supply.

Reliability and Testing

Reliability stresses multiply with die size and packaging complexity. Thermal cycling can delaminate interfaces on large chips. Panel substrates suffer from mechanical fatigue as noted [30†L168-L177] . Thus extensive qualification is needed: testing chips under temperature/humidity, accelerated life tests on solder joints and carriers, and scanning for electromigration in large metal nets. Multi-die packages require ensuring each bond (TSV, microbump, or bridge) is robust. Known-good-die (KGD) testing improves yield for chiplets: each tile is tested at wafer probe before assembly. Wafer-scale chips can be tested via probe stations that cover large areas or by dividing the wafer. Burn-in and redundancy strategies (as Cerebras uses) help reliability: allowing faulty units to be masked keeps system uptime

high [6†L62-L68] . In data-centers, hot-swap of modules and system-level mirroring (RAID for compute) are also used.

Deployment in Data Centres

- **Cooling infrastructure:** AI accelerators exceeding ~500 W/TDP typically need liquid cooling. Data centers housing wafer-scale boards (≥ 20 kW each [6†L38-L43]) must install chilled-water racks or immersion tanks. Floor plans may cluster these high-power nodes together with independent cooling loops. Redundancy (N+1 chillers, dual pump loops) is essential to avoid single-point failures.
- **Power delivery:** Racks hosting many high-TDP chips require high-capacity PDUs and power distribution (e.g. 48–380 V DC rails to racks). Advanced power modules (low voltage drop, dynamic current sharing) are needed. Communication overhead (for power management) can be large: NVLink/CXL provide not only data but also side-band power control interfaces.
- **Interconnect fabrics:** Large chips often drive the need for new rack-scale networks. E.g., Cerebras’s CS-4 system uses “wafer links” between multiple WSEs to compose larger fabrics. NVIDIA DGX or AMD Infinity systems use NVLink/NVSwitch meshes inside rack. CXL adoption will lead to new memory and accelerator fabric switches in racks. Optical backplanes are being developed for >1 m board-to-board links (preventing copper density bottlenecks).
- **Redundancy & maintenance:** Having one *huge* chip per node can be a risk: a single die defect could offline tens of TFLOPS. Designers mitigate by using multiple chips per server or enabling partial failover. For example, AMD APU cards like MI300A have 4 dies; if one fails, the card still functions (at reduced capability). At the system level, clustering multiple nodes and checkpointing software lets jobs migrate off failed hardware. Graphcore and Cerebras emphasize “software-transparent” handling of faults within their chips, but service teams still must consider replacement policies for large boards.
- **Rack design:** Traditional 19” racks may need modification: some wafer-scale engines (CS-3/CS-4) occupy full rack width/height. Weight and EMI are also considerations. Cable and airflow management must adapt (liquid pipes, fiber harnesses). The industry is moving toward pre-integrated AI racks (e.g. Lenovo, Supermicro chassis for 2–4 GPUs) which could extend to exascale AI appliances (cerebras style).

Comparative Analysis

The table below contrasts four approaches to scaling chip area: **Monolithic** (single-die on 300 mm wafer), **Chiplet/MCM** (multi-chip modules, 2.5D), **Wafer-scale** (stitched full wafer), and **Panel-scale**. Each row shows key metrics and trade-offs.

Approach	Max Die Size	Latency	Bandwidth	Yield Impact	Relative Cost	Manufacturability
Monolithic	~26×33 mm (≈900 mm ²) (limit)	Lowest (single-chip, internal)	Moderate (on-chip bus, e.g. HBM)	Worst: yield $\propto e^{-a}$ (with A) [8†L41-L44]	Low (widely mature)	Very mature CMO process

Approach	Max Die Size	Latency	Bandwidth	Yield Impact	Relative Cost	Manufacturability
Chiplet	Scales by tiling (virtually unlimited)	Medium (off-die, but package-bound)	High (many interfaces: Infinity Fabric, UCIe)	Better (smaller die yields high, ~2–10× more chips/wafer 【8†L59-L68】)	Medium-High (packaging costs)	Requires advanced packaging (EMIB, CoWoS)
Wafer-scale	Full wafer (~714 mm diameter ≈73,000 mm ²)	Low (on-wafer network, like on-chip)	Extreme (10–100× TB/s on-chip 【1†L92-L100】)	Variable: <i>handled by redundancy</i> 【6†L62-L68】 ; raw yield ≈0 without it	Very high (custom fab, huge cost)	Very challenging (yield, design, cooling)
Panel-scale	Panels ≈300×300 to 600×600 mm (90,000–360,000 mm ²)	Low/Medium (on-interposer network)	Very high (panel interposers can carry many dies, large aggregate I/O)	Currently <i>very high risk</i> : glass/wafers cracks 【30†L168-L177】	Unknown (development cost high; volume uncertain)	Immature (pilot, requires new fab/tools)

Notes: *Monolithic* chips today top ~800 mm² (Blackwell). *Chiplets* combine multiple dies (e.g. AMD's 4×370 mm² tiles). *Wafer-scale* (Cerebras) is ~46k mm² currently. *Panel* projects like 310×310 mm (≈96k mm²) are in early trials 【30†L145-L154】 . Latency is lowest within a single die, increases off-chip (e.g. chiplet hop ~ns, board-level tens of ns). Bandwidth: monolithic HBM might be ~6–8 TB/s (shared memory); wafer and panel can reach 10s–100s TB/s aggregate 【1†L92-L100】 . Yield impact: monolithic falls off steeply with area 【8†L41-L44】 , chiplets split risk, wafer relies on redundancy 【6†L62-L68】 , panels currently need breakthroughs in substrate reliability 【30†L168-L177】 . Cost rank (L→H): monolithic < chiplet < wafer < panel (development). Manufacturability: monolithic chip scaling is standard (but limited in size), chiplets need advanced R&D (but many suppliers), wafer/panel need end-to-end new flows. Time-to-market: chiplets are mainstream now, wafer-scale is in niche production, panel-scale is emerging late-decade.

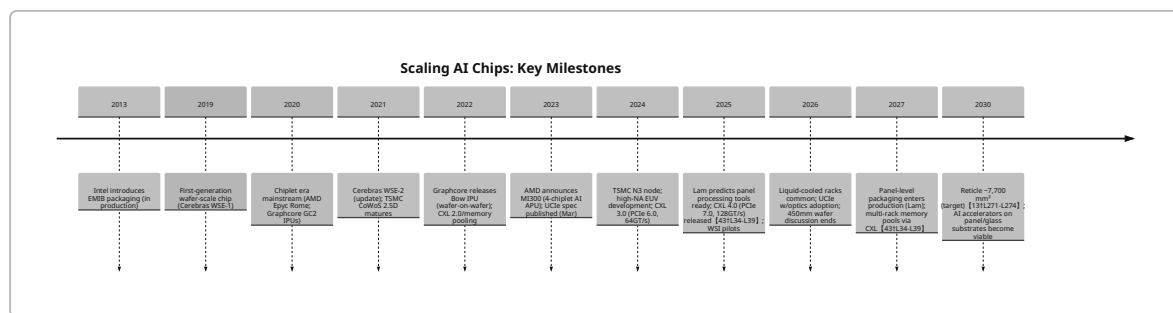
Recommendations

- **Short term (next 1–2 years):** Focus on **chiplet/2.5D** approaches for large AI chips. This leverages existing fab and packaging technology, improving yield and time-to-market. Use silicon interposers or EMIB bridges with HBM for high bandwidth. Adopt UCIe as it matures (standardized chiplet interface) to simplify multi-vendor integration 【24†L99-L104】 . Prepare for CXL-4 adoption (2026–27) to enable memory pooling across systems 【43†L34-L39】 . Implement robust cooling (liquid) and multi-phase power delivery for any >300 W modules 【6†L38-L43】 【36†L219-L227】 .
- **Medium term (3–5 years):** Invest in **wafer-scale and panel-scale R&D**. Develop design tools and redundancies for wafer-scale chips (mesh networks, sparing) as AI pushes beyond chiplet limits 【6†L62-L68】 【1†L71-L79】 . Collaborate on panel infrastructure: support industry consortia, prototype panel processing for fan-out packaging 【13†L271-L274】 . Build supply

chains for large glass substrates (Absolics/SKC) and panel litho tools (Nikon, ASML X2).
 Standardize interfaces: UCIe with photonics to link across dies and packages with pJ/bit efficiency
 【24†L47-L54】 【24†L119-L128】 .

- **Design trade-offs:** For a given product, weigh **latency vs. complexity**. Monolithic yields best latency/bandwidth but is size-limited and low-yield. Chiplets increase latency slightly (board-level hops) but dramatically improve yield and modularity 【8†L41-L44】 . Wafer-scale gives minimal latency and massive bandwidth 【1†L92-L100】 , but requires novel architectures (asynchronous, resilient mesh) and specialized cooling 【6†L38-L43】 . Hybrid solutions may dominate: e.g. chiplets interconnected by silicon photonics or CXL for near-off-chip coherence.
- **Deployment best practices:** Data-center operators should design racks with **modularity and redundancy**. Ensure power infrastructure can support >50 kW per rack for multiple high-end chips. Adopt liquid cooling now in AI clusters and be ready to upgrade valves/monitors for next-gen chips 【6†L38-L43】 . Plan networks that combine PCIe/CXL fabrics with high-speed optical fabrics for scaling. Encourage vendors to expose defect maps of large devices (if possible) for cluster scheduling.

Mermaid Timeline



Sources: Authoritative industry and academic references have been cited throughout. Key figures and tables are drawn from recent white papers (Cerebras, TSMC), research (USENIX 【1†L71-L80】), and technical press (Semiconductor Engineering, EETimes) to ensure up-to-date accuracy.